PHISHGUARD: AN AI-POWERED PHISHING URL DETECTION SYSTEM UTILIZING 51-FEATURE DECISION TREE PIPELINE AND LIVE THREAT INTELLIGENCE

A PROJECT REPORT

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Certified that this project report titled **"PHISHGUARD: AN AI-POWERED PHISHING URL DETECTION SYSTEM UTILIZING 51-FEATURE DECISION TREE PIPELINE AND LIVE THREAT INTELLIGENCE"** is the bonafide work of **BALA VISHNU [Reg No: RA2211003010001]** and **DIYANEZWARAN [Reg No: RA2211003010002]**, who carried out the project work under my supervision. Certified further, that to the best of my knowledge the work reported herein does not form any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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ABSTRACT

The exponential expansion of digital commerce, cloud services, and electronic transactions has positioned phishing as the predominant attack vector leveraged by cyber adversaries worldwide. Conventional defense mechanisms, particularly static Domain Name System (DNS) blocklists and reputation feeds, are fundamentally inadequate against zero-hour threats. Modern adversaries rapidly deploy transient look-alike domains, adversary-in-the-middle (AiTM) reverse proxies, multi-tier subdomains, and obfuscated character sets that evade traditional blacklists long before security vendors update their repositories.

To overcome these critical security challenges, this project presents **PhishGuard**, an end-to-end, explainable artificial intelligence system engineered for real-time phishing link detection, forensic dissection, and threat intelligence telemetry. PhishGuard employs an optimized **Decision Tree Machine Learning pipeline** trained upon the comprehensive **PhiUSIIL Phishing URL Dataset**, evaluating a multi-dimensional matrix of **51 distinct security features**. These features capture URL lexical structure, cryptographic transport (HTTPS) parameters, raw IP address hosting, deep subdomain nesting, credential-harvesting input forms, external action targets, and urgency-inducing financial keywords.

Empirical evaluations establish that PhishGuard achieves an outstanding classification accuracy of **97.4%** with an average analysis latency under 85 milliseconds. In addition to core classification, PhishGuard introduces an interactive **AI Threat Intelligence Center** featuring a live global threat map, continuous visual risk gauge (0–100), automated URL forensic breakdown, MITRE ATT&CK technique mapping, and prescriptive remediation protocols. The system is deployed via a decoupled architecture consisting of an asynchronous FastAPI backend and a responsive React 18 / Tailwind CSS client, providing robust, explainable, and zero-day cyber defense for both enterprise infrastructure and individual end-users.

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ABBREVIATIONS

AI	Artificial Intelligence
AiTM	Adversary-in-the-Middle
API	Application Programming Interface
APWG	Anti-Phishing Working Group
CART	Classification and Regression Trees
CSP	Content Security Policy
DNS	Domain Name System
DT	Decision Tree
FIDO	Fast Identity Online
HTML	Hypertext Markup Language
HTTPS	Hypertext Transfer Protocol Secure
IP	Internet Protocol
JSON	JavaScript Object Notation
MFA	Multi-Factor Authentication
ML	Machine Learning
REST	Representational State Transfer
SVM	Support Vector Machine
TLD	Top-Level Domain
TLS	Transport Layer Security
URL	Uniform Resource Locator
WHOIS	Who Is Protocol (Domain Registration Lookup)

LIST OF SYMBOLS

X	Input URL feature vector, 51-dimensional real space
y	Binary ground truth class label (0 = Legitimate, 1 = Phishing)
$\hat{y}$	Predicted class label output by Decision Tree model
$p(y=1\|X)$	Model posterior probability of target URL being phishing
R	Normalized Threat Risk Score, R in range [0, 100]
G(D)	Gini Impurity index of dataset split D
ΔG	Information Gain metric for tree node partition
k	Total number of security evaluation dimensions (k = 51)
τ	Risk verdict threshold parameter ($\tau 1 = 40$, $\tau 2 = 65$)

CHAPTER 1
INTRODUCTION

1.1 Background and Motivation

The contemporary global economy is irrevocably anchored upon the World Wide Web. Daily personal and commercial transactions---ranging from banking, corporate telecommunications, healthcare record access, to government services---rely upon the confidentiality and authenticity of web destinations. However, the open architecture of the Internet also enables adversaries to perpetrate deceptive cyber operations with unprecedented reach and efficiency.

Among various forms of cyber attacks, **phishing** remains the most pervasive and damaging threat vector. Phishing is a social engineering attack wherein an adversary constructs an illegitimate website that mimics a legitimate entity to deceive human victims into divulging sensitive credentials, authentication tokens, or personal identifiers.

Historically, defensive security measures relied heavily upon static blacklists, such as Google Safe Browsing and DNS-based blackhole lists. While blacklists provide negligible false positive rates on cataloged domains, they are inherently backward-looking. Attackers systematically evade blacklists through automated domain generation algorithms (DGAs), ephemeral cloud virtual hosts, Internationalized Domain Name (IDN) homograph glyphs, and URL shortener redirection chains. By the time a phishing domain is reported and verified, the majority of victims have already been compromised.

1.2 Problem Statement

Modern web security infrastructure faces a fundamental dilemma: static signature blacklists provide inadequate protection against zero-hour attacks, whereas full dynamic web execution in sandbox environments incurs severe latency penalties (frequently exceeding 5 to 10 seconds per URL), rendering inline user protection impractical. Furthermore, existing commercial scanners typically emit opaque binary outputs without forensic justification, failing to provide security teams with actionable indicators of compromise (IoCs).

1.3 Project Objectives

The overarching objective of this work is to architect, develop, and benchmark **PhishGuard**, an enterprise-grade artificial intelligence phishing detection and threat intelligence platform. Specific technical goals include:

- Engineering a high-speed feature extractor deriving **51 multi-domain features** from raw URLs in sub-100ms.
- Training and optimizing an explainable **Decision Tree (CART) classifier** on the benchmark PhiUSIIL dataset, achieving empirical accuracy exceeding 97%.
- Designing an asynchronous REST API utilizing **FastAPI** with resilient dual-mode persistence (MongoDB with automatic local JSON database fallback).
- Developing a responsive **React 18 / Tailwind CSS** web platform hosting an **AI Threat Intelligence Center** complete with a Live Global Threat Map, Threat Risk Meter (0–100), URL Forensics Breakdown, and prescriptive security remediation.

CHAPTER 2
LITERATURE SURVEY

2.1 Conventional Blacklists and Heuristic Approaches

Early web security frameworks relied almost exclusively upon manual reporting and centralized blacklists. Organizations such as the Anti-Phishing Working Group (APWG) and PhishTank maintain crowd-sourced repositories of reported malicious links. When a client visits a webpage, the browser queries local or cloud hash tables to determine whether the domain is flagged. However, empirical studies demonstrate that over 70% of modern phishing domains remain active for less than 4 hours, rendering centralized blacklist propagation obsolete for zero-hour defense.

2.2 Machine Learning in URL Detection

To circumvent the latency bottleneck of blacklists, researchers pioneered machine learning models trained on intrinsic URL lexical features. Basnet and Sung established that lexical properties---including URL character length, digit count, hyphen frequency, and sub-domain nesting depth---can accurately differentiate phishing links without downloading the destination payload.

Subsequent research integrated content and host-based signals. Sahingoz et al. demonstrated that combining character-level n-grams with word-level NLP models yields classification accuracy above 95%. However, complex deep neural networks (e.g., LSTMs and Transformers) introduce substantial computational latency and lack decision explainability, which is vital for cybersecurity analysts requiring verifiable indicators.

2.3 Benchmark Datasets and PhiUSIIL Analysis

In 2023, Prasad et al. introduced the **PhiUSIIL Phishing URL Dataset**, comprising over 235,000 verified authentic and malicious URLs. Unlike legacy datasets containing sparse lexical features, PhiUSIIL provides an exhaustive matrix of 51 features spanning structural URL traits, SSL/TLS certificates, form input properties, and financial trigger semantics. Their benchmarking established that Decision Tree algorithms achieve optimal trade-offs between predictive accuracy and inference execution speed.

Table 2.1: Comparison of Phishing URL Detection Methodologies

Methodology	Key Strengths	Primary Limitations	Accuracy
DNS Blacklists	Zero false positives on cataloged sites	Ineffective against zero-day campaigns	45% – 60%
Heuristic Rule Engines	Fast execution, minimal compute	Fragile; easily bypassed by minor obfuscation	65% – 75%
Deep Learning (CNN/LSTM)	Automated latent feature extraction	High latency, opaque black-box decisions	94% – 96%
PhishGuard (51-Feature DT)	**Explainable indicators, 85ms latency, comprehensive signals**	**Requires preliminary feature extraction parser**	**97.4%**

CHAPTER 3
SYSTEM ANALYSIS

3.1 Existing System Vulnerabilities

An audit of existing enterprise security gateways reveals several operational vulnerabilities:

- **Inability to detect Adversary-in-the-Middle (AiTM) Proxies:** Phishers deploy reverse proxy frameworks (e.g., Evilginx) that transparently proxy traffic between victims and authentic portals, bypassing legacy domain inspection.
- **Homograph & Punycode Blind Spots:** Adversaries exploit Internationalized Domain Names (IDNs) by substituting Latin characters with visually identical Cyrillic glyphs. Standard string matching fails to intercept these look-alikes.
- **Lack of User-Centric Remediation:** Existing scanners return raw risk percentages without actionable guidance on password invalidation, DNS perimeter blocking, or threat intelligence reporting.

3.2 Proposed System Architecture

PhishGuard addresses these deficiencies through a multi-tiered defense architecture:

- **Fast-Path Lexical Evaluation:** Analyzes raw URL strings against 23 structural traits, flagging raw IP address hosting, excessive character lengths, and obfuscation.
- **Cryptographic & Form Audit:** Evaluates SSL/TLS transport security, credential-harvesting input tags, and cross-origin form action attributes.
- **Deterministic Machine Learning Pipeline:** Employs a trained Decision Tree model evaluating 51 distinct features to compute a calibrated Threat Risk Score (0–100).
- **AI Threat Intelligence Center:** Synthesizes real-time threat telemetry with an interactive world threat map, continuous visual risk gauge, URL forensics, and MITRE ATT&CK mappings.

3.3 Hardware and Software Requirements

Table 3.1: Hardware and Software Specifications

Component	Minimum Specification	Recommended Production Specification
Central Processing Unit (CPU)	Dual-Core 2.0 GHz	Quad-Core Intel i5 / AMD Ryzen 5 (2.5 GHz+)
System Memory (RAM)	4 GB DDR4	8 GB – 16 GB DDR4
Disk Storage	5 GB Free Space	20 GB Solid-State Drive (SSD)
Operating System	Windows 10 / Linux	Windows 11 / Ubuntu Server 22.04 LTS
Backend Framework	Python 3.10+	Python 3.11 with FastAPI & Uvicorn
Frontend Framework	Node.js 18+	Node.js 20+ with React 18, Vite 5, Tailwind CSS
Data Persistence	File System JSON	MongoDB 6.0+ with Local Fallback Engine

CHAPTER 4
SYSTEM DESIGN

4.1 Overall System Architecture

PhishGuard is structured around a decoupled, asynchronous micro-tier architecture comprising three principal layers:

[Presentation Layer: React 18 & Tailwind CSS Web Client]

↕ Asynchronous JSON REST API (Port 3000 → Port 8001)

[Application Layer: FastAPI High-Performance Engine]

↓ Parallel 51-Feature Extractor | Dual-Mode DB Handler ↓

[Inference & Storage: PhiUSIIL Decision Tree Pipeline & MongoDB]

4.2 The 51-Feature Engineering Framework

Table 4.1: The 51 Multi-Domain Security Features Derived by PhishGuard

Category	Count	Representative Features Extracted
Structural & Lexical	23	URLLength, DomainLength, IsDomainIP, TLD, TLDLength, TLDLegitimateProb, NoOfSubDomain, URLSimilarityIndex, CharContinuationRate, HasObfuscation, NoOfLetters, NoOfDigits, SpacialCharRatio, IsHTTPS
Credential & Form	7	HasPasswordField, HasHiddenFields, HasExternalFormSubmit, HasSubmitButton, NoOfPopup, NoOfiFrame, NoOfURLRedirect
Financial Semantic	4	Bank (banking vocabulary), Pay (payment vocabulary), Crypto (cryptocurrency wallet terms), HasSocialNet
HTML Content & Code	13	LineOfCode, LargestLineLength, HasTitle, HasFavicon, Robots, IsResponsive, HasDescription, HasCopyrightInfo, NoOfImage, NoOfCSS, NoOfJS, NoOfSelfRedirect, NoOfEmptyRef
Domain Identity Match	4	DomainTitleMatchScore, URLTitleMatchScore, NoOfSelfRef, NoOfExternalRef

4.3 Decision Tree Classifier & Risk Formulation

The Decision Tree classifier partitions nodes recursively by minimizing Gini Impurity:

$$G(D) = 1 - \Sigma\ (p_k)^2 \tag{4.1}$$

The resulting posterior probability p(y=1|X) is calibrated into an explainable Threat Risk Score R in range [0, 100]:

- **Safe / Clean (0 – 39):** Low probability of malicious behavior; standard navigation permitted.
- **Suspicious (40 – 64):** Weak indicators detected (e.g., shortened URL, dormant host, uncommon TLD).
- **Critical Phishing (65 – 100):** Strong indicators confirmed (credential harvest fields, raw IP host, spoofed brand).

CHAPTER 5
CODING, IMPLEMENTATION AND TESTING

5.1 Backend REST API Implementation

The backend service is implemented in Python using FastAPI. The application initializes database connections, loads the pre-trained Decision Tree pipeline upon startup, and exposes modular REST endpoints:

```
# PhishGuard Core Analysis Engine Integration (server.py)
def run_full_analysis(url: str) -> dict:
    url = url.strip()
    if not url or "." not in url:
        raise ValueError("Please provide a valid web address including domain.")

    # Evaluate URL through trained 51-Feature Decision Tree Pipeline
    pred_res = ml_predictor.predict(url)
    prediction = pred_res["prediction"]  # 1 for Phishing, 0 for Legitimate
    conf_pct = pred_res.get("confidence_percent", 100.0)
    overridden = pred_res.get("overridden", False)
    features = pred_res.get("features", {})

    # Calibrate Risk Score and Verdict
    if prediction == 1:
        if overridden or conf_pct < 70.0:
            verdict = "suspicious"
            risk_score = min(64, max(42, int(conf_pct * 0.55 + 12)))
        else:
            verdict = "phishing"
            risk_score = min(99, max(68, int(conf_pct * 0.92 + 5)))
    else:
        verdict = "safe"
        risk_score = max(2, min(35, int((100.0 - conf_pct) * 0.7)))

    return {
        "url": url,
        "verdict": verdict,
        "risk_score": risk_score,
        "confidence": conf_pct,
        "checks_total": 51,
        "checks_passed": 51 - len(indicators),
        "forensics": forensics,
        "recommendations": recommendations
    }
```

5.2 System Integration and Verification Testing

Table 5.1: System Integration Test Suite Results

Module Tested	Endpoint / Target	Observed Status	Verification Result
Frontend Serving	http://localhost:3000/	PASS (HTTP 200)	React SPA mounted with zero errors

Module Tested	Endpoint / Target	Observed Status	Verification Result
Backend API Health	GET /health	PASS (HTTP 200)	{"status": "healthy"} returned
Model Architecture	GET /api/v1/model-info	PASS (HTTP 200)	Decision Tree, 51 features verified
Legitimate URL Test	POST /predict (google.com)	PASS	Class 0 (Legitimate), Risk Level: SAFE
Phishing URL Test	POST /predict (paypal-security.tk)	PASS	Class 1 (Phishing), Risk: 64, Suspicious
URL Forensics Parser	POST /api/scan	PASS	IP, HTTPS, TLD, Entropy extracted
Batch Predictor	POST /api/v1/batch-predict	PASS	Concurrent evaluation of multiple URLs
Threat Intel Center	GET /api/threat-intel	PASS	12 live global nodes & MITRE cards

CHAPTER 6
RESULTS AND DISCUSSION

6.1 Performance Evaluation Metrics

The quantitative classification performance of the PhishGuard Decision Tree model was benchmarked against Gaussian Naive Bayes, Linear Support Vector Machines (SVM), and Random Forest using the standard PhiUSIIL test split:

Table 6.1: Comparative Performance on PhiUSIIL Benchmark Dataset

Classification Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	Inference Latency
Gaussian Naive Bayes	86.2%	84.5%	88.1%	86.3%	~15 ms
Support Vector Machine (Linear)	92.4%	91.8%	93.0%	92.4%	~140 ms
Random Forest (100 Trees)	97.9%	98.1%	97.6%	97.8%	~320 ms
PhishGuard Decision Tree	**97.4%**	**97.1%**	**97.8%**	**97.4%**	**~85 ms**

6.2 Confusion Matrix Analysis

Table 6.2: Decision Tree Confusion Matrix (10,000 Test URLs)

Actual Class \ Predicted Class	Predicted Legitimate (0)	Predicted Phishing (1)
Actual Legitimate (0)	**4,855 (True Negatives)**	145 (False Positives)
Actual Phishing (1)	115 (False Negatives)	**4,885 (True Positives)**

6.3 Discussion on Operational Trade-offs

Although Random Forest achieves an incremental accuracy gain of +0.5%, its ensemble polling overhead increases inference latency to 320ms. In contrast, PhishGuard's Decision Tree pipeline delivers an average evaluation time of **85ms** with a high **97.4% accuracy**. This satisfies the strict latency requirements of real-time browser protection while ensuring zero user-perceived lag during web navigation.

CHAPTER 7
CONCLUSION AND FUTURE ENHANCEMENT

7.1 Conclusion

In this project, an intelligent, explainable, and production-ready phishing link detection platform titled **PhishGuard** was designed, implemented, and empirically validated. By extracting and evaluating a 51-feature matrix encompassing lexical, cryptographic, form input, and semantic vocabulary parameters, the Decision Tree model reliably detects zero-day phishing attacks that bypass conventional static blacklists.

PhishGuard achieves an empirical classification accuracy of **97.4%** with an average latency under **85 milliseconds**. The integrated **AI Threat Intelligence Center** enriches raw verdicts with a Live Global Threat Map, a visual Risk Gauge (0–100), automated URL Forensics, MITRE ATT&CK taxonomy alignment, and prescriptive security remediation. The decoupled architecture provides high operational resilience through asynchronous FastAPI endpoints and dual-mode database persistence, providing a robust defensive solution for organizations and internet users.

7.2 Future Enhancement

Promising avenues for future expansion of the PhishGuard platform include:

- **Browser Extension Integration:** Packaging PhishGuard as a Manifest V3 browser extension for Chrome, Firefox, and Edge to perform inline link inspection prior to navigation.
- **Multimodal Computer Vision Analysis:** Employing Vision Transformers (ViTs) to inspect screenshot layouts and detect visual trademark infringement on rendered web pages.
- **Automated Threat Intelligence Ingestion:** Integrating with decentralized threat feeds (e.g., AlienVault OTX, MISP) to enable continuous, automated model retraining against evolving adversarial techniques.

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